SOFTWARE DEVELOPMENT SEPM

Assignment - 3

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Q.1. To understand DevOps: Principle, Practices and devops Engineer Role and responsibilities.

DevOps is a combination of development (Dev) and operations (ops). It is a culture, set of practices and tools that improves collaboration between software developers and IT operations teams. The main goal of devops is to deliver software faster, with better quality and with continuous improvement.

a Principles of DevOps:

① Collaboration and communication
Development, testing and operation teams work together throughout the software lifecycle.

② Automation: Repetitive tasks like testing, building deployment and monitoring are automated to save time & reduce errors.

③ Continuous Integration (CI):
Developers frequency merge code into a shared repository where builds and tests run automatically.

④ Continuous Delivery / continuous Deployment (CI):
Software updates are released quickly and reliably through automated pipelines.

⑤ Monitoring and feedback: Applications and infrastructure are continuously monitored to detect issues and improve performance

⑥ Customer centric approach:
Focus is on delivering value to users faster with frequent update and improvement

⑦ Infrastructure as Code (IaC): Focus is on delivering value to users faster with frequent

updates and improvement.

(7) ~~Infrastructure as~~

* DevOps Practices
  - ↳ continuous Integration (CI)
  - ↳ continuous delivery (CD)
  - ↳ continuous Deployment
  - ↳ version control
  - ↳ Automated Testing
  - ↳ configuration management
  - ↳ containerization
  - ↳ Orchestration
  - ↳ Monitoring and logging

* Role of a devOps Engineer

A devOps Engineer is responsible for bringing the gap between development and operation teams. They automate workflows manage infrastructure and ensure reliable software delivery.
They work with developers, tests, security teams and cloud teams to maintain smooth release cycles.

* Responsibilities of a DevOps Engineer
  - ↳ Build and Release management: create automated build and deployment pipelines:
  - ↳ CI/CD pipeline setup: use tools like Jenkins, github action or github CI/CD
  - ↳ Infrastructure management: provision and maintain servers, cloud resources &

Automation : automate testing, deployment backups and repetitive tasks.
Monitoring and incident response.
Security Integration (DevSecOps) : Include security checks in pipeline and infrastructure
Collaboration : coordinate between developers, QA, and operations teams.
Performance optimization : Improve scalability, reliability etc
Documentation : Maintain runbooks, development guides, and architecture documents.